Branch/Class/Batch: AIDS/TE/A1
Student Roll no.: A16
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EXPERIMENT NUMBER: 8

AIM: To design and simulate VLANs on switch/router using any network simulator (e.g. Cisco Packet Tracer, GNS3)

OBJECTIVES: To create VLANs, assign switch ports to different VLANs, and verify network connectivity using the ping command..

THEORY:

1. What is a VLAN?

1. What is a VLAN?

VLAN (Virtual Local Area Network) is a logical way of dividing one physical network into multiple separate networks.

For example, a single switch can be configured with:

- VLAN 10 → HR department
- VLAN 20 → Sales department
- VLAN 30 → IT department

Even though all the computers may be connected to the same physical switch, devices in different VLANs are logically separated.

2. What is the purpose of creating VLANs in a network?

VLANs are mainly created to:

- **Improve security:** Devices in one VLAN are isolated from devices in another VLAN.
- **Reduce broadcast traffic:** Broadcasts are normally limited to the VLAN, reducing unnecessary network traffic.
- **Organize the network:** Users can be grouped logically by department, function, or purpose rather than physical location.
- **Improve network management:** Administrators can manage different groups of users separately.
- **Provide flexibility:** A user can remain in the same logical network even if they move to a different physical location, provided the switch ports are configured

appropriately.

Example:
If HR and Sales computers are connected to the same physical switch, VLANs can keep their network traffic separated.

3. What is the difference between a LAN and a VLAN?

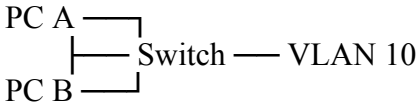
LAN	VLAN
Local Area Network	**Virtual Local Area Network**
A network covering a limited physical area, such as an office or building.	A logically separated network created within a physical network.
Usually refers to the physical/local network.	Provides logical segmentation of that network.
Devices can be connected through switches, cables, Wi-Fi, etc.	VLANs are configured on managed network devices such as switches.
Separation is often based on physical network structure.	Separation is based on logical configuration.

4. What is the difference between communication within the same VLAN and communication between different VLANs?

Communication within the same VLAN

Devices belonging to the **same VLAN** can communicate directly through the switch, assuming normal IP addressing and security settings allow it.

For example:



If both PCs are in VLAN 10, their traffic can be switched directly between them.

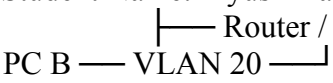
Communication between different VLANs

Devices in **different VLANs cannot communicate directly through a normal Layer 2 switch**.

For example:



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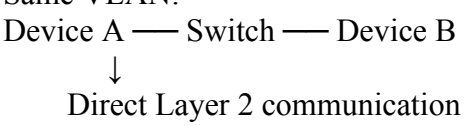


Communication between VLAN 10 and VLAN 20 requires **inter-VLAN routing**, which is performed by a **router or Layer 3 switch**.

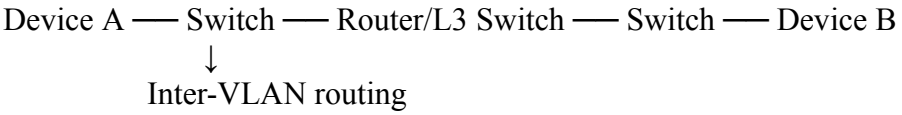
The router/Layer 3 switch can also apply rules such as **ACLs (Access Control Lists)** to control which VLANs are allowed to communicate.

In short

Same VLAN:



Different VLANs:



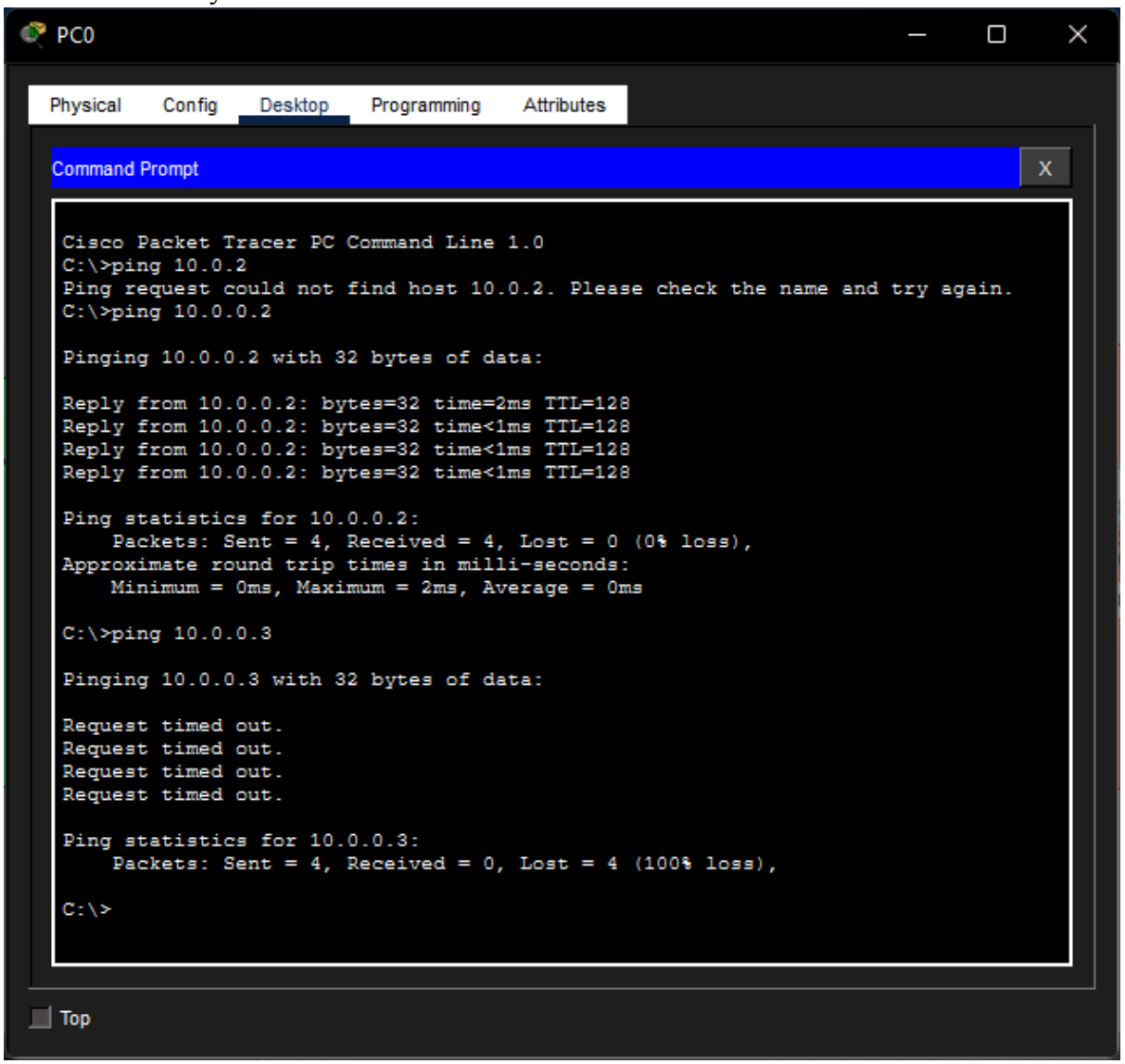
RESOURCES REQUIRED:
Switch
4PCs

PROCEDURES:

1. Create a topology with one switch and four PCs in Cisco Packet Tracer and connect them using appropriate cables.
2. Assign IP addresses from the `10.0.0.x` network to all four PCs, e.g., `10.0.0.1`–`10.0.0.4` with subnet mask `255.0.0.0`.
3. Test connectivity between different PCs using the `ping` command. Verify that all PCs can communicate.
4. Create two VLANs on the switch, e.g., VLAN 10 – Office and VLAN 20 – Home.
5. Assign PC0 and PC1 to VLAN 10, and PC2 and PC3 to VLAN 20.
6. Test connectivity again using the `ping` command.
7. Verify that PCs within the same VLAN can communicate, while PCs belonging to different VLANs cannot communicate through a Layer-2 switch.

RESULT:

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Switch0

Physical Config CLI Attributes

```
% Invalid input detected at '^' marker.

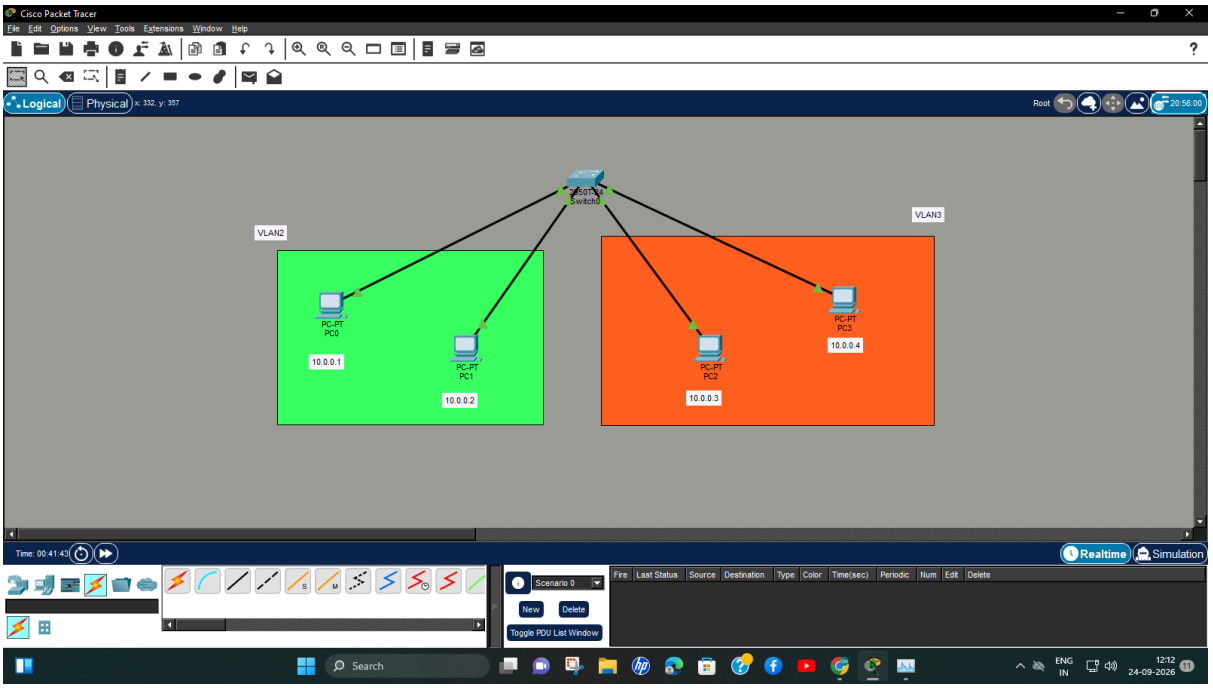
Switch(config-vlan)#name office
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name home
Switch(config-vlan)#exit
Switch(config)#vlan2
                  ^
% Invalid input detected at '^' marker.

Switch(config)#interface fastethernet0/1
Switch(config-if)#switchport acc
% Incomplete command.
Switch(config-if)#switchport access vlan2
                                    ^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastethernet0/3
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#interface fastethernet0/2
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastethernet0/4
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#
```

Copy Paste

Top



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CONCULSION:

The VLAN topology was successfully designed and simulated in Cisco Packet Tracer. Two VLANs, VLAN 10 (Office) and VLAN 20 (Home), were created and the PCs were assigned to their respective VLANs. The ping test confirmed that PCs within the same VLAN could communicate with each other, while PCs belonging to different VLANs could not communicate through the Layer-2 switch. Thus, VLANs provide effective network segmentation, improved security, and reduced broadcast traffic.